50. Parallel Data Capture Unit (PDC)

50.1 Overview

This MCU includes a single parallel data capture unit (PDC).
The PDC has the function of communicating with the external I/O devices including image sensors, and specifically transferring parallel data such as an image output from the external I/O devices via the DTC or DMAC to the on-chip RAM and external address spaces (the CS and SDRAM areas).

Table 50.1 PDC Specifications

Item	Description
Capture Range	• Desired amounts of parallel data within the following ranges in the vertical and horizontal directions. Vertical direction: 1 to 4095 lines Horizontal direction: 4 to 4095 bytes
Parallel data transfer clock (PIXCLK)	• Operating frequency: 1 to 27 MHz*1
Interrupt Sources	• Receive data ready • Frame end • Overrun • Underrun • Error in the setting for the number of lines • Error in setting for the number of bytes per line
Startup of DTC/DMAC	• The receive data ready interrupt is capable of starting the DTC or DMAC
Parallel data transfer clock output (PCKO)	• Operating frequency: 1 to 30 MHz*2 • Clock source: Peripheral module clock B (PCLKB) • Frequency division ratio: Selectable from 2, 4, 6, 8, 10, 12, 14, and 16.
Others	• PDC reset function • Selectable active sense for the VSYNC and HSYNC signals • Monitoring of the VSYNC and HSYNC signals • Endian selection
Reducing Power Consumption	• The PDC can be placed in the module stop state.
Internal Bus Interface	• The PDC is connected with internal peripheral bus 3

Note 1. The frequency of the parallel data transfer clock (PIXCLK) should be equal to or less than 0.6 × PCLKB (peripheral module clock).
Note 2. The operating frequency is 30 MHz when that of the peripheral module clock B (PCLKB) is 60 MHz and the frequency division ratio is 2.

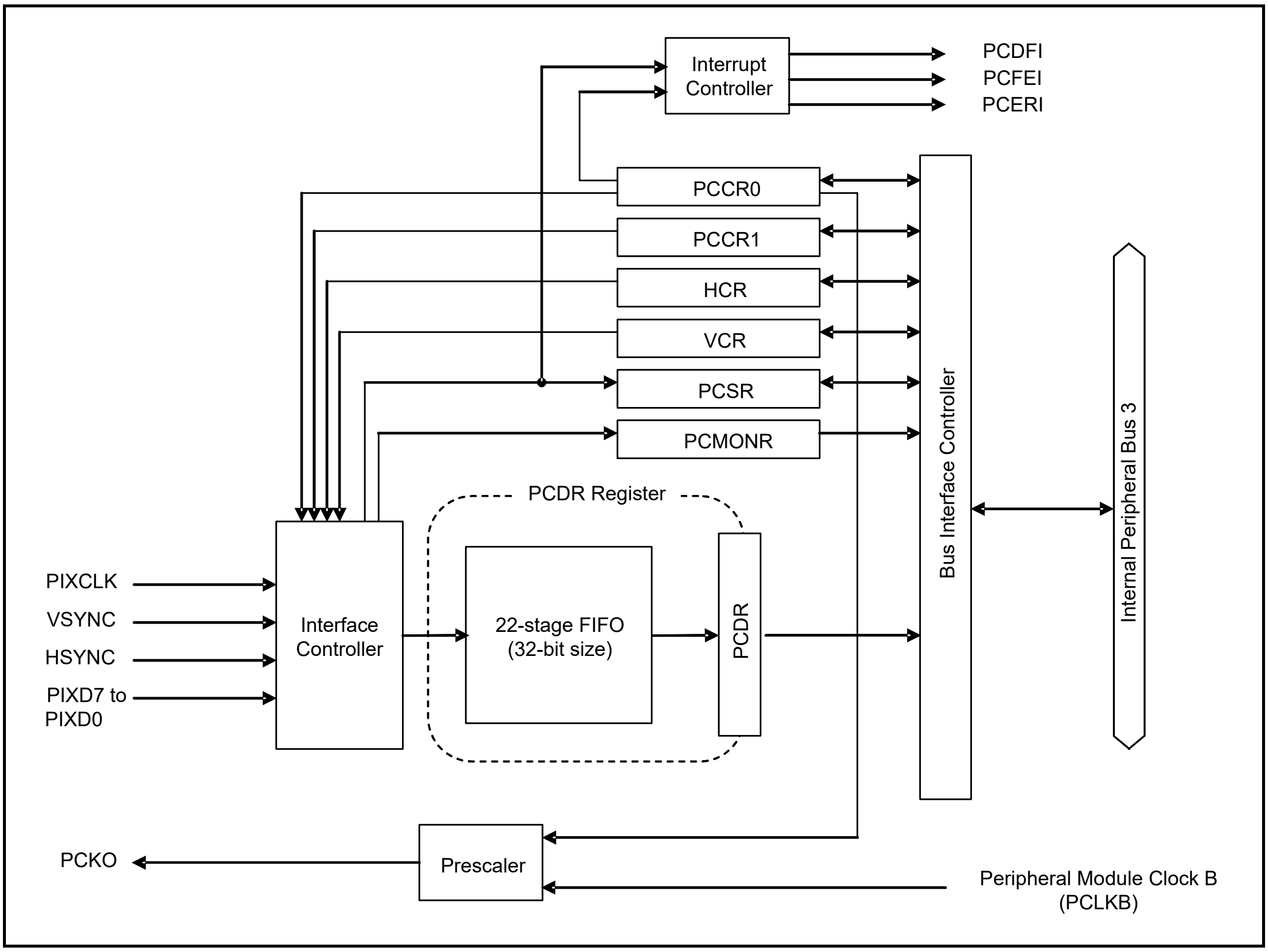


Figure 50.1 Block Diagram of the PDC

Table 50.2 Input and Output Pins of the PDC

Pin Name	Input/Output	Description
PIXCLK	Input	Parallel data transfer clock
VSYNC	Input	Vertical synchronization signal
HSYNC	Input	Horizontal synchronization signal
PIXD7 to PIXD0	Input	8-bit data
PCKO	Output	Output of the parallel data transfer clock

50.2 Register Descriptions

50.2.1 PDC Control Register 0 (PCCR0)

Address(es): 000A 0500h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	EDS	PCKDIV[2:0]			PCKOE	HERIE	VERIE	UDRIE	OVIE	FEIE	DFIE	PRST	HPS	VPS	PCKE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PCKE	PIXCLK Input Enable	0: PIXCLK input is disabled. 1: PIXCLK input is enabled.	R/W
b1	VPS	VSYNC Signal Polarity Select	0: VSYNC signal is active high. 1: VSYNC signal is active low.	R/W
b2	HPS	HSYNC Signal Polarity Select	0: HSYNC signal is active high. 1: HSYNC signal is active low.	R/W
b3	PRST	PDC Reset	0: PDC reset is not applied. 1: PDC is reset.	R/(W) *1
b4	DFIE	Receive Data Ready Interrupt Enable	0: Generation of receive data ready interrupt requests is disabled. 1: Generation of receive data ready interrupt requests is enabled.	R/W
b5	FEIE	Frame End Interrupt Enable	0: Generation of frame end interrupt requests is disabled. 1: Generation of frame end interrupt requests is enabled.	R/W
b6	OVIE	Overrun Interrupt Enable	0: Generation of overrun interrupt requests is disabled. 1: Generation of overrun interrupt requests is enabled.	R/W
b7	UDRIE	Underrun Interrupt Enable	0: Generation of underrun interrupt requests is disabled. 1: Generation of underrun interrupt requests is enabled.	R/W
b8	VERIE	Vertical Line Number Setting Error Interrupt Enable	0: Generation of vertical line number setting error interrupt requests is disabled. 1: Generation of vertical line number setting error interrupt requests is enabled.	R/W
b9	HERIE	Horizontal Byte Number Setting Error Interrupt Enable	0: Generation of horizontal byte number setting error interrupt requests is disabled. 1: Generation of horizontal byte number setting error interrupt requests is enabled.	R/W
b10	PCKOE	PCKO Output Enable	0: PCKO output is disabled (fixed to the high level) 1: PCKO output is enabled.	R/W
b13 to b11	PCKDIV[2:0]	PCKO Frequency Division Ratio Select	b13 b11 0 0 0: PCKO/2 0 0 1: PCKO/4 0 1 0: PCKO/6 0 1 1: PCKO/8 1 0 0: PCKO/10 1 0 1: PCKO/12 1 1 0: PCKO/14 1 1 1: PCKO/16	R/W
b14	EDS	Endian Select	0: Little endian 1: Big endian	R/W
b31 to b15	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

Note 1. Only 1 can be written to this bit.

The PCCR0 register should only be set while the PCE bit in the PCCR1 register is 0.

PCKE Bit (PIXCLK Input Enable)

This bit enables and disables the input through the PIXCLK pin.
Set this bit to 1 before enabling reception. After enabling input through the PIXCLK pin, use the PRST bit to initialize the PDC.
When setting this bit to 0, do so after disabling reception operations.

VPS Bit (VSYNC Signal Polarity Select)

This bit selects the active sense of the VSYNC signal.

HPS Bit (HSYNC Signal Polarity Select)

This bit selects the active sense of the HSYNC signal.

PRST Bit (PDC Reset)

This bit initializes the internal status of the PDC and the target registers of the PDC reset. Refer to section 50.3.11, Reset State for the target registers.
The PDC should be reset after setting the PCKE bit to 1.
When 1 is written to the PRST bit, initialization is started in synchronization with the PIXCLK. After completion of the initialization, the PRST bit is automatically cleared to 0. When the PDC is reset, ensure that the PIXCLK pin has an input signal. Also, after 1 has been written to the PRST bit, do not proceed to the next step until verifying that the bit has returned to 0.
In the case of consecutive PDC resets, wait for at least one cycle of PIXCLK after verifying that the PRST bit has returned to 0.

DFIE Bit (Receive Data Ready Interrupt Enable)

This bit enables and disables the generation of receive data ready interrupt requests.

FEIE Bit (Frame End Interrupt Enable)

This bit enables and disables the generation of frame end interrupt requests.

OVIE Bit (Overrun Interrupt Enable)

This bit enables and disables the generation of overrun interrupt requests.

UDRIE Bit (Underrun Interrupt Enable)

This bit enables and disables the generation of underrun interrupt requests.

VERIE Bit (Vertical Line Number Setting Error Interrupt Enable)

This bit enables and disables the generation of vertical line number setting error interrupt requests.

HERIE Bit (Horizontal Byte Number Setting Error Interrupt Enable)

This bit enables and disables the generation of horizontal byte number setting error interrupt requests.

PCKOE Bit (PCKO Output Enable)

This bit enables and disables an output from PCKO.
When the PCKOE bit is cleared to 0 during low output of PCKO, it may cause high output at the time of clearing, resulting in spoiling the duty factor.

PCKDIV[2:0] Bits (PCKO Frequency Division Ratio Select)

This bit selects the frequency division ratio of the PCKO.
The PCKO output is a clock signal derived by dividing the PCLKB clock signal by the value from 2 to 16 corresponding to the setting of the PCKDIV[2:0] bits.
The operating frequency should be set in accord with that of PCLKB to a value within the range from 1 to 30 MHz.

EDS Bit (Endian Select)

This bit selects an endian for the captured data.

50.2.2 PDC Control Register 1 (PCCR1)

Address(es): 000A 0504h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	PCE	PDC Operation Enable	0: Operations for reception are disabled. 1: Operations for reception are enabled.	R/W
b31 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

PCE Bit (PDC Operation Enable)

This bit enables and disables operations for reception.
When the PCE bit is set to 1 during assertion of the VSYNC signal, the PDC starts operations for reception from the next effective edge of the VSYNC signal.
The PCE bit should only be cleared to 0 while operations for reception or continued reception are stopped, including the frame end interrupt and so on. Regarding operations for continued reception, refer to section 50.3.6, Operations for Continued Reception at the Time of Frame End.

50.2.3 PDC Status Register (PCSR)

Address(es): 000A 0508h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	HERF	VERF	UDRF	OVRF	FEF	FEMPF	FBSY
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0

Bit	Symbol	Bit Name	Description	R/W
b0	FBSY	Frame Busy Flag	0: Operations for reception are stopped. 1: Operations for reception are ongoing.	R
b1	FEMPF	FIFO Empty Flag	0: FIFO is not empty. 1: FIFO is empty.	R
b2	FEF	Frame End Flag	0: Frame end has not been generated. 1: Frame end has been generated.	R/(W) *1
b3	OVRF	Overrun Flag	0: FIFO overrun has not been generated. 1: FIFO overrun has been generated.	R/(W) *1
b4	UDRF	Underrun Flag	0: Underrun has not been generated. 1: Underrun has been generated.	R/(W) *1
b5	VERF	Vertical Line Number Setting Error Flag	0: Vertical line number setting error has not been generated. 1: Vertical line number setting error has been generated.	R/(W) *1
b6	HERF	Horizontal Byte Number Setting Error Flag	0: Horizontal byte number setting error has not been generated. 1: Horizontal byte number setting error has been generated.	R/(W) *1
b31 to b7	—	Reserved	These bits are read as 0. The write value should be 0.	R

Note 1. 0 can only be written to these bits to clear these flags after they have been read out as 1.

FBSY Flag (Frame Busy Flag)

This flag indicates the state of PDC operations.

[Setting Condition]

- Detection of the effective edge of the VSYNC signal after the enabling of operations for reception.

[Clearing Conditions]

- Reception of one-frame of data according to the settings of the VCR and HCR registers being completed.*1
- When an error (overrun, underrun, vertical line number setting error, or horizontal byte number setting error) occurs
- When the PCCR1.PCE bit is 0

Note 1. This flag is 0 during operations for continued reception.

FEMPF Flag (FIFO Empty Flag)

This flag indicates the state of the FIFO.

This flag indicates the state of the FIFO at the time when a vertical line number setting error or a horizontal byte number setting error occurs and is 0 following an overrun and undefined following an underrun.

[Setting Conditions]*1

- Reading of the PCDR register while the FIFO is empty.

- Detection of an effective edge of the VSYNC signal.
- The PDC being reset.

[Clearing Condition]*1

- Storage of the data captured in the FIFO.

Note 1. This flag is undefined following an underrun, and is set to either 0 or 1.

FEF Flag (Frame End Flag)

This flag indicates the end of a frame.

[Setting Condition]

- Reception of one frame of data in accord with the settings of the VCR and HCR registers.*1

[Clearing Conditions]

- The PDC being reset.
- Writing of 0 to the bit after it has been read as 1.

Note 1. In the case of operations for continued reception, this flag will be 1 after their completion.

OVRF Flag (Overrun Flag)

This flag indicates an overrun.

[Setting Condition]

- Data for reception arriving while the FIFO is full.

[Clearing Conditions]

- The PDC being reset.
- Writing of 0 to the bit after it has been read as 1.

UDRF Flag (Underrun Flag)

This flag indicates an underrun.

[Setting Condition]

- Reading of the PCDR register while the FIFO is empty.

[Clearing Conditions]

- The PDC being reset.
- Writing of 0 to the bit after it has been read as 1.

VERF Flag (Vertical Line Number Setting Error Flag)

This flag indicates an error in the setting for the number of lines.

[Setting Condition]

- The VSYNC signal is negated because fewer lines have been captured than the value in the VCR register.

[Clearing Conditions]

- The PDC being reset.
- Writing of 0 to the bit after it has been read as 1.

HERF Flag (Horizontal Byte Number Setting Error Flag)

This flag indicates an error in the setting for the number of bytes in a line.

[Setting Condition]

- The HSYNC signal is negated because fewer bytes in a line have been captured than the value in the HCR register.

[Clearing Conditions]

- The PDC being reset.
- Writing of 0 to the bit after it has been read as 1.

50.2.4 PDC Pin Monitor Register (PCMONR)

Address(es): 000A 050Ch

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	—	—	—	—	—	—	—	HSYNC	VSYNC
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	VSYNC	VSYNC Signal Status Flag	0: VSYNC signal is at the low level. 1: VSYNC signal is at the high level.	R
b1	HSYNC	HSYNC Signal Status Flag	0: HSYNC signal is at the low level. 1: HSYNC signal is at the high level.	R
b31 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R

VSYNC Flag (VSYNC Signal Status Flag)

This flag indicates the state of the VSYNC signal.

HSYNC Flag (HSYNC Signal Status Flag)

This flag indicates the state of the HSYNC signal.

50.2.5 PDC Receive Data Register (PCDR)

Address(es): 000A 0510h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

The PDC includes a 32-bit, 22-stage FIFO buffer for the storage of captured data. The FIFO buffer is mapped to the 4-byte PCDR register, and captured data are read from this register in 4-byte units. The receive data ready flag is set for every 32 bytes of received data, and this also leads to a receive data ready interrupt if the DFIE bit in the PCCR0 register is set to 1. When a receive data ready interrupt is generated, read the PCDR register eight times. Figure 50.2 shows a schematic view of the PCDR register.

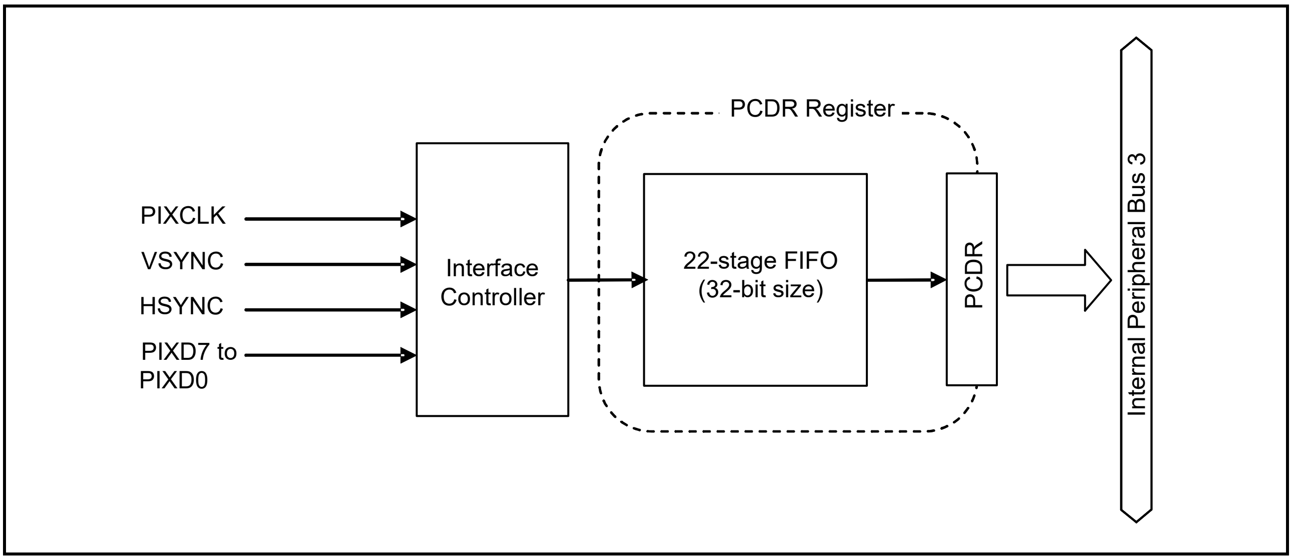


Figure 50.2 Schematic View of the PCDR Register

For the format of the captured data, either big or little endian can be selected by the EDS bit of the PCCR0 register. Figure 50.3 shows the arrangements of data according to the endian formats.

Reading of the PCDR register	Big Endian 31			0	Little Endian 31			0
First time	0 byte	1 byte	2 byte	3 byte	3 byte	2 byte	1 byte	0 byte
Second time	4 byte	5 byte	6 byte	7 byte	7 byte	6 byte	5 byte	4 byte
. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .	. . .
n[*1] time	4(n-1) byte	4(n-1) + 1 byte	4(n-1) + 2 byte	4(n-1) + 3 byte	4(n-1) + 3 byte	4(n-1) + 2 byte	4(n-1) + 1 byte	4(n-1) byte

Note 1. n = Number of reading of the PCDR register

Figure 50.3 Endian Formats

50.2.6 Vertical Capture Register (VCR)

Address(es): 000A 0514h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	VSZ[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	VST[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b11 to b0	VST[11:0]	Vertical Capture Start Line Position	Number of the line where capture is to start.	R/W
b15 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b27 to b16	VSZ[11:0]	Vertical Capture Size	Number of lines to be captured.	R/W
b31 to b28	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

For the relationship between the VCR register setting and the captured range, refer to section 50.3.3, Settings of the VCR and HCR Registers and the Captured Range.
The VCR register should be set while the PCE bit in the PCCR1 register is 0.

VST[11:0] Bit (Vertical Capture Start Line Position)

These bits set the number of the line where capture is to start.
To set the first line, these bits should be 000h; to set the 4095th line, they should be FFEh.
The setting of the VST[11:0] bits should be within the range from 000h to FFEh and, in combination with that of the VSZ[11:0] bits, satisfy the following relation:
Setting range of the VST[11:0] bits: $1 \leq \text{VST[11:0]} + \text{VSZ[11:0]} \leq \text{FFFh}$.

VSZ[11:0] Bit (Vertical Capture Size)

These bits set the number of lines to be captured.
To set one line, these bits should be 001h; to set 4095 lines, they should be FFFh.
The setting of the VSZ[11:0] bits should be within the range from 001h to FFFh and, in combination with that of the VST[11:0] bits, satisfy the following relation:
Setting range of the VSZ[11:0] bits: $1 \leq \text{VST[11:0]} + \text{VSZ[11:0]} \leq \text{FFFh}$.

50.2.7 Horizontal Capture Register (HCR)

Address(es): 000A 0518h

	b31	b30	b29	b28	b27	b26	b25	b24	b23	b22	b21	b20	b19	b18	b17	b16
	—	—	—	—	HSZ[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

	b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	HST[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b11 to b0	HST[11:0]	Horizontal Capture Start Byte Position	Horizontal position in bytes where capture is to start.	R/W
b15 to b12	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b27 to b16	HSZ[11:0]	Horizontal Capture Size	Number of bytes to be captured horizontally.	R/W
b31 to b28	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

For the relationship between the HCR register setting and the captured range, refer to section 50.3.3, Settings of the VCR and HCR Registers and the Captured Range.
The HCR register should be set while the PCE bit in the PCCR1 register is 0.

HST[11:0] Bit (Horizontal Capture Start Byte Position)

These bits set the horizontal position in bytes where capture is to start.
To set the first byte, these bits should be 000h; to set the 4092th byte, they should be FFBh.
The setting of the HST[11:0] bits should be within the range from 000h to FFBh and, in combination with that of the HSZ[11:0] bits, satisfy the following relation:
Setting range of the HST[11:0] bits: 4 ≤ HST[11:0] + HSZ[11:0] ≤ FFFh.

HSZ[11:0] Bit (Horizontal Capture Size)

These bits set the number of bytes to be captured per line.
To set four bytes, these bits should be 004h; to set 4095 bytes, they should be FFFh.
The setting of the HSZ[11:0] bits should be within the range from 004h to FFFh and, in combination with that of the HST[11:0] bits, satisfy the following relation:
Setting range of the HSZ[11:0] bits: 4 ≤ HST[11:0] + HSZ[11:0] ≤ FFFh.

50.3 Operation

50.3.1 Transfer Formats

The PDC supports the four transfer formats shown in Figure 50.4 to Figure 50.7. The format is determined by the setting of the VPS and HPS bits in the PCCR0 register.

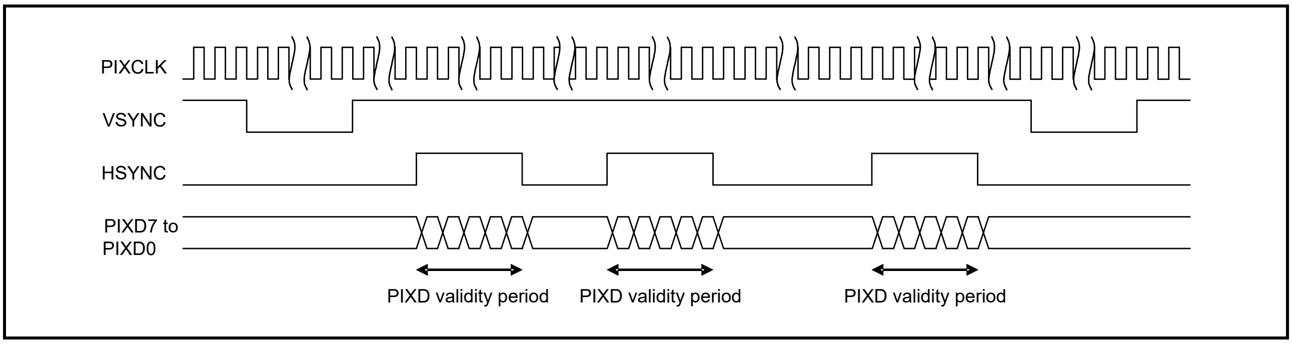


Figure 50.4 PDC Transfer Format (for VPS=0, HPS=0)

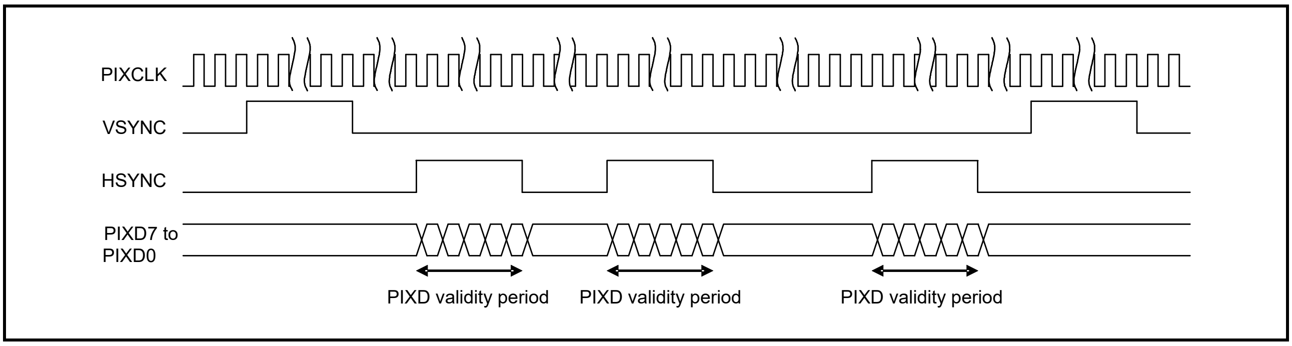


Figure 50.5 PDC Transfer Format (for VPS=1, HPS=0)

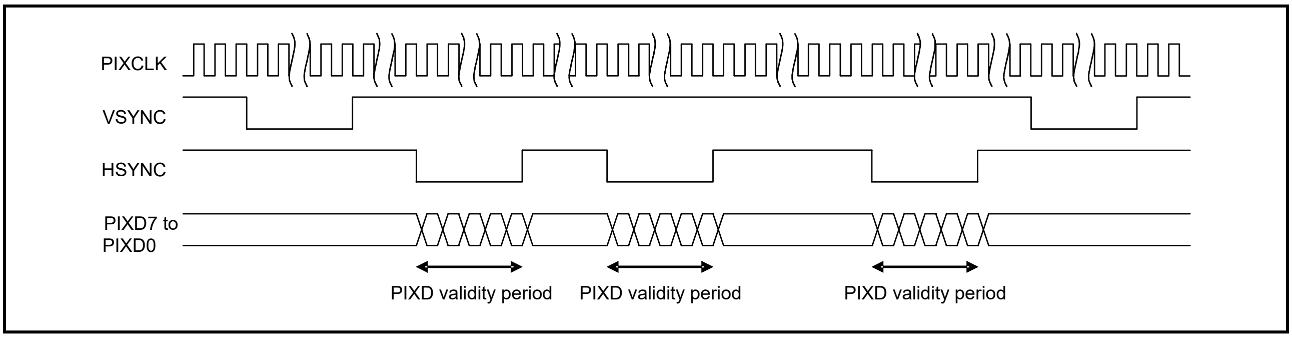


Figure 50.6 PDC Transfer Format (for VPS=0, HPS=1)

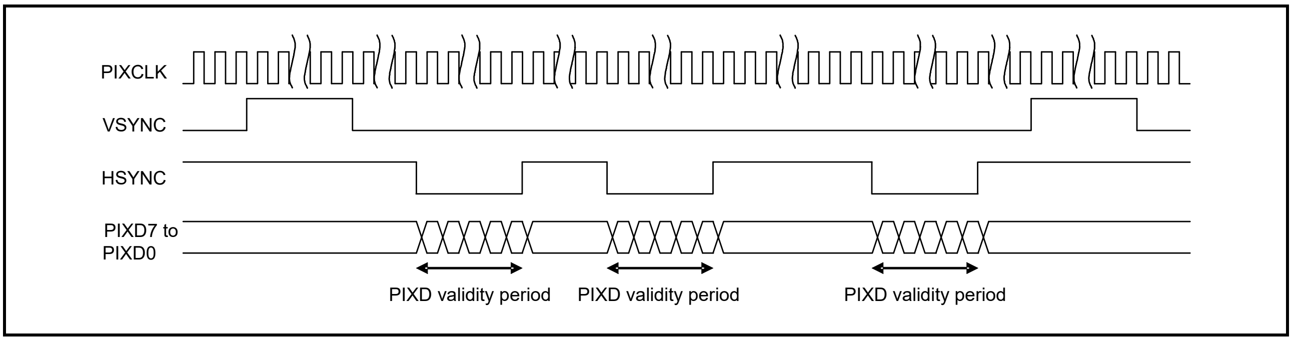


Figure 50.7 PDC Transfer Format (for VPS=1, HPS=1)

50.3.2 Transfer Timing

Figure 50.8 and Table 50.3 show the timing of transfer by the PDC.

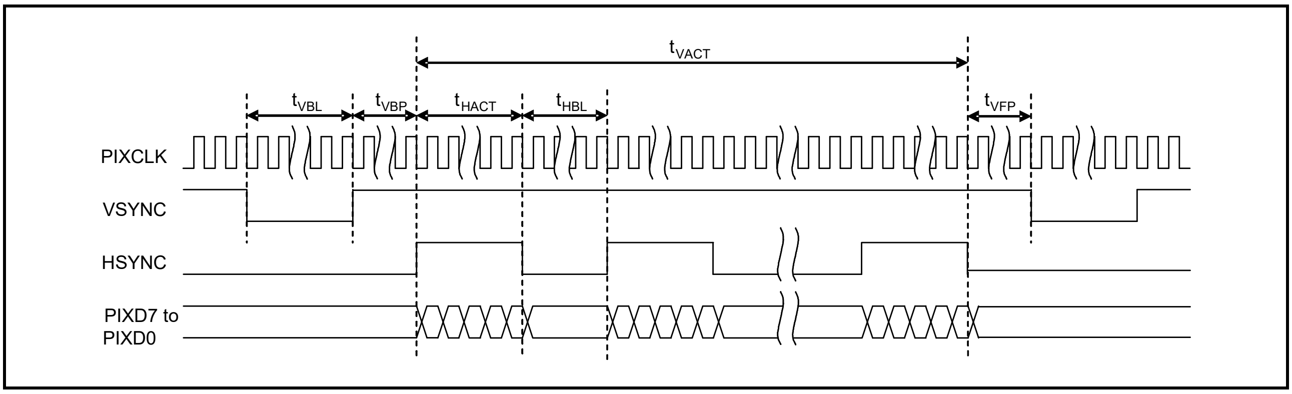


Figure 50.8 PDC Transfer Timing

Table 50.3 PDC Transfer Timing

Item	Symbol	Min*1	Max	Unit
Vertical Blanking Period	t_{VBL}	128	—	PIXCLK
Vertical Backporch	t_{VBP}	10	—	PIXCLK
Horizontal Valid Period	t_{HACT}	4	4095	PIXCLK
Horizontal Blanking Period	t_{HBL}	128	—	PIXCLK
Vertical Frontporch	t_{VFP}	10	—	PIXCLK
Vertical Valid Period	t_{VACT}	1	4095	Line

Note 1. The minimum values are the lowest of which this MCU circuit is capable. They are not values which guarantee the avoidance of overruns, vertical line number setting errors, or horizontal byte number setting errors.

50.3.3 Settings of the VCR and HCR Registers and the Captured Range

Figure 50.9 and Figure 50.10 show the relationship between the settings of the VCR and HCR registers and the captured range.

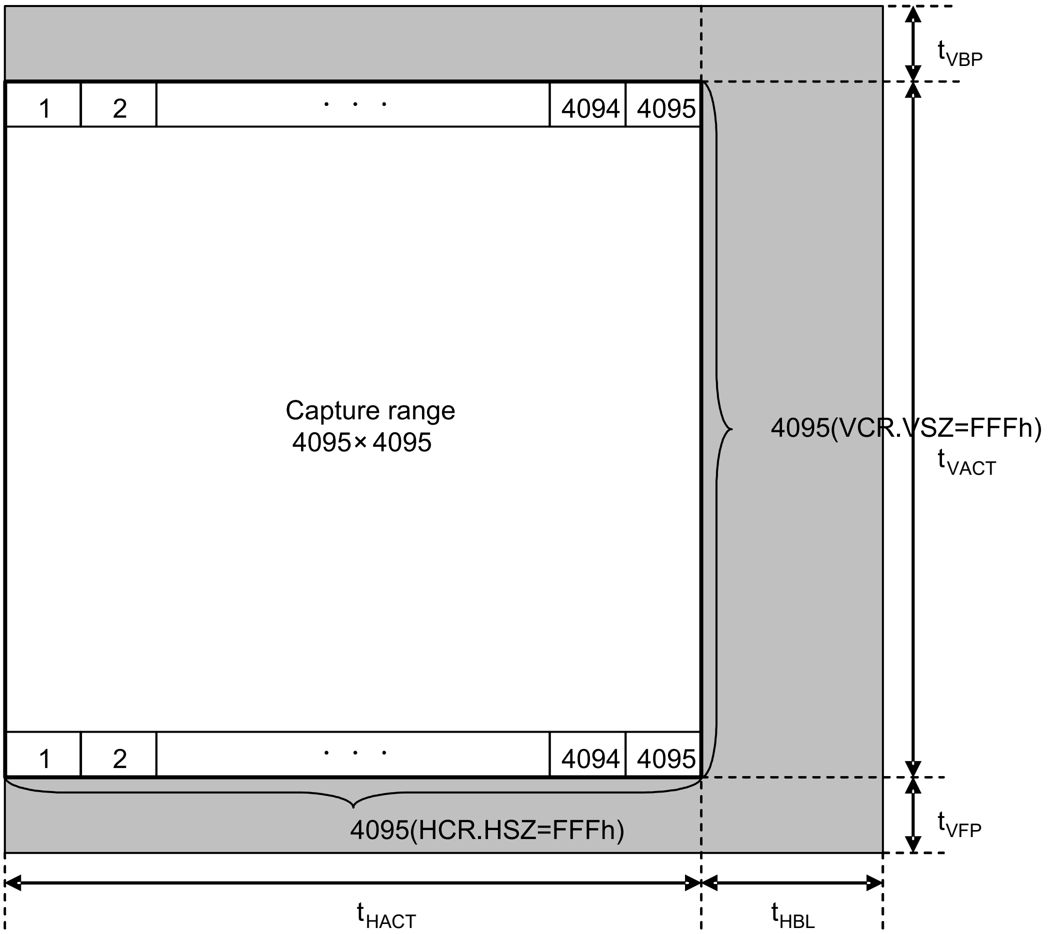


Figure 50.9 Settings of the VCR and HCR Registers and the Captured Range (VCR = 0FFF 0000h, HCR = 0FFF 0000h)

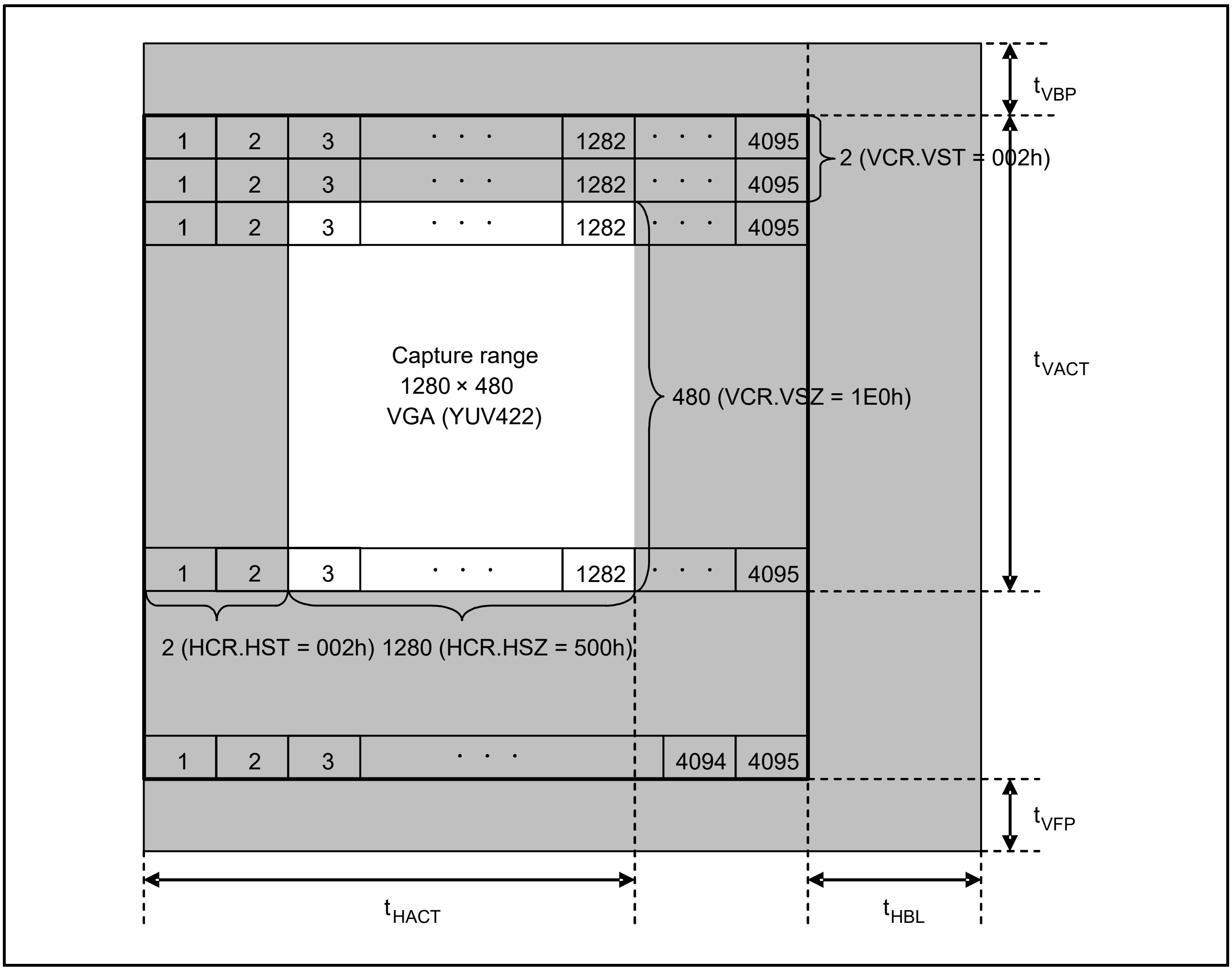


Figure 50.10 Settings of the VCR and HCR Registers and the Captured Range (VCR = 01E0 0002h, HCR = 0500 0002h)

50.3.4 Operations for Reception

Figure 50.11 shows an example of operations for reception when receive data ready interrupts (to start up the DTC or DMAC) and frame end interrupts are in use.

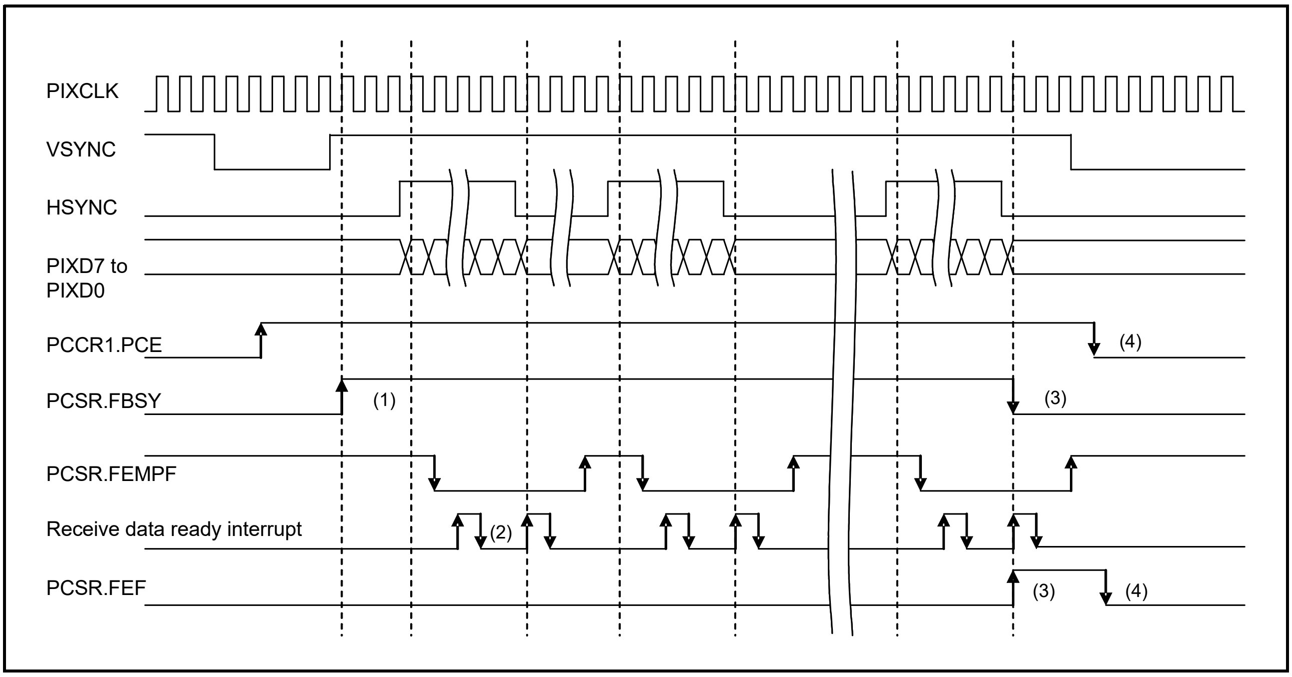


Figure 50.11 Example of Operations for Reception

The actual operations at the times indicated by (1), (2), (3), and (4) in Figure 50.11 are described below.

(1) When an effective edge of the VSYNC signal is detected after the PCE bit in the PCCR1 register is set to 1, the FEMPF flag in the PCSR register is set to 1 and the FIFO is initialized. Concurrently, the FBSY flag in the PCSR register is set to 1 and operations for reception start.

(2) When data within the range for capture set in the VCR and HCR registers is received, the data are stored in the FIFO. The PDC generates a receive data ready interrupt every time it receives 32 bytes of data, and the interrupt starts transfer of the captured data by the DTC or DMAC to the on-chip RAM or an external address space. The FIFO is likely to overrun if reading of the PCDR register takes more time than reception of the data. Check the OVRF flag in the PCSR register to verify an overrun.

(3) After reception of the last byte of data is completed, the FBSY flag in the PCSR register is cleared to 0 and the FEF flag in the PCSR register is set to 1 so that a receive data ready interrupt and frame end interrupt are generated.

(4) The FEMPF flag in the PCSR register is polled by the frame end interrupt, after which the completion of data transfer by the DTC or DMAC should be verified. After the PCE bit in the PCCR1 register is cleared to 0, the FEF flag in the PCSR register is also cleared to 0, and the reception of one frame of data is complete.
If an underrun occurs while the PCSR.FEMPF flag is being polled, setting of the PCSR.FEMPF flag may not proceed. Accordingly, also check the PCSR.UDRF flag during polling, and if an underrun occurs, run appropriate error processing.

If the FEF flag in the PCSR register has been set to 1 before the PCE bit in the PCCR1 register is set to 1, effective edges of the VSYNC signal are not detected and operations for reception will not be started. Clear the FEF flag in the PCSR register to 0 to start operations for data reception.

50.3.5 Operation during the Horizontal Blanking Period

If the horizontal blanking period begins but the number of the received data bytes has not reached 32 bytes since the previous receive data ready, the count of the number of received data bytes is retained and carried over to the next horizontal valid period. Figure 50.12 shows an example of operation during the horizontal blanking period.

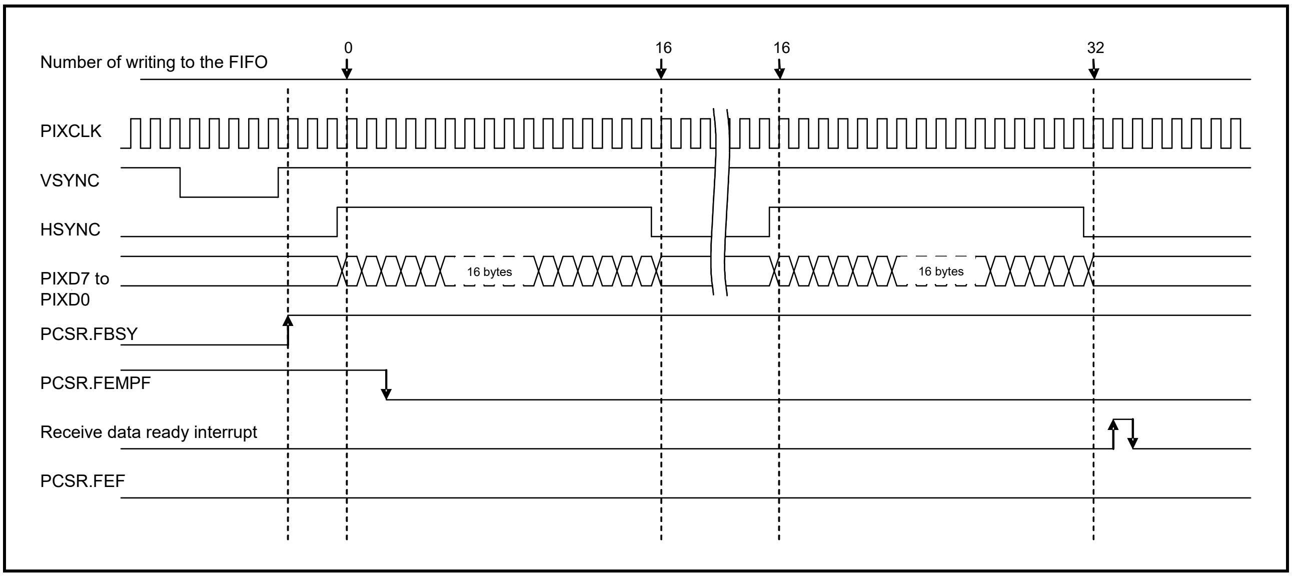


Figure 50.12 Example of Operation during the Horizontal Blanking Period

50.3.6 Operations for Continued Reception at the Time of Frame End

When the last of the data have been received but the number of bytes of data received since the previous receive data ready has not reached 32, the PDC continues to receive data (hereinafter referred to as “continued reception”) until the number reaches 32. After that, it generates a receive data ready and a frame end. The PIXCLK should be input during operations for continued reception. If the data stored in the FIFO are read out during operations for continued reception, the values read are undefined. Figure 50.13 shows an example of operations at the time of frame end.

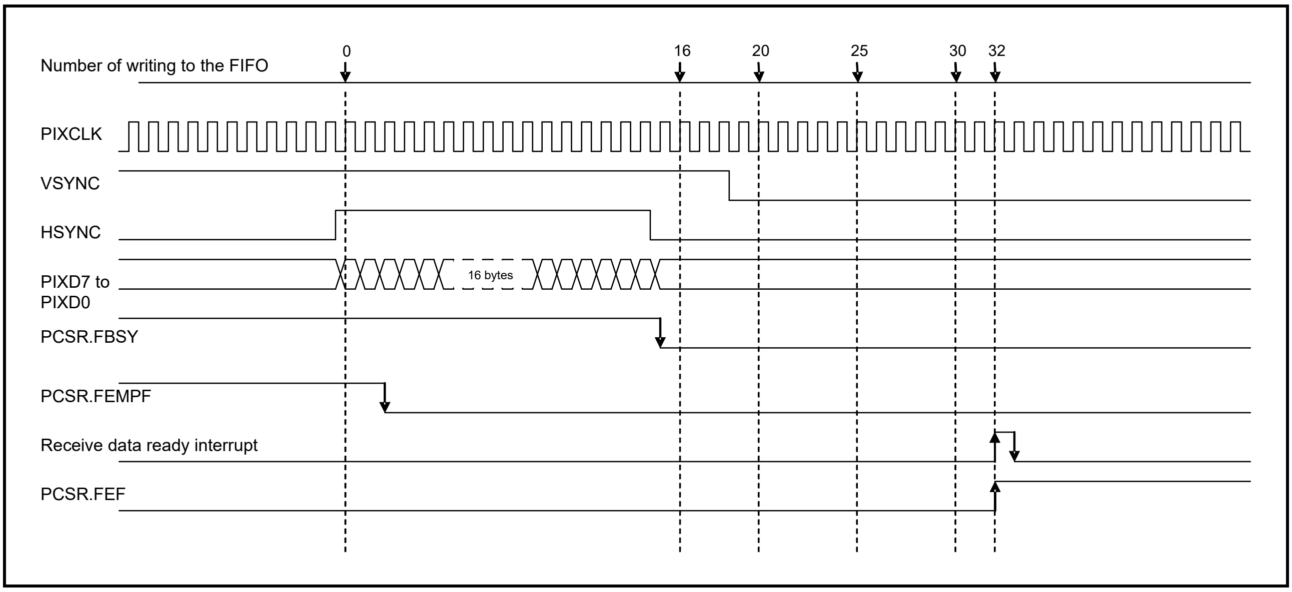


Figure 50.13 Example of Continued Reception at the Time of Frame End

50.3.7 Error Detection

The PDC has functions for error detection so that the software can apply control in response to errors during operations for reception. Table 50.4 summarizes the conditions under which each type of error is detected and the interrupt flags set in response.

Table 50.4 Error Detection

Error Factor	Conditions of Error Detection	Interrupt Flag	Example of Operation
Overrun	Data for reception arriving while the FIFO is full.*1	PCSR.OVRF	Figure 50.14
Underrun	The PCDR register being read while the FIFO is empty.	PCSR.UDRF	Figure 50.15
Vertical line number setting error	Negation of the VSYNC signal when the number of captured lines is less than the value set in the VCR register.	PCSR.VERF	Figure 50.16
Horizontal byte number setting error	Negation of the HSYNC signal when the number of bytes captured in a line is less than the value set in the HCR register.	PCSR.HERF	Figure 50.17

Note 1. This includes data reception during operations for continued reception.

When an error is detected, the PDC sets the corresponding interrupt flag to 1 to stop operations for reception. While the interrupt flag is set to 1, the PDC does not detect effective edges of the VSYNC signal and does not start operations for reception. Clear all error source interrupt flags to 0 to start operations for reception.
In addition, when an error occurs, data stored in the FIFO is disabled.

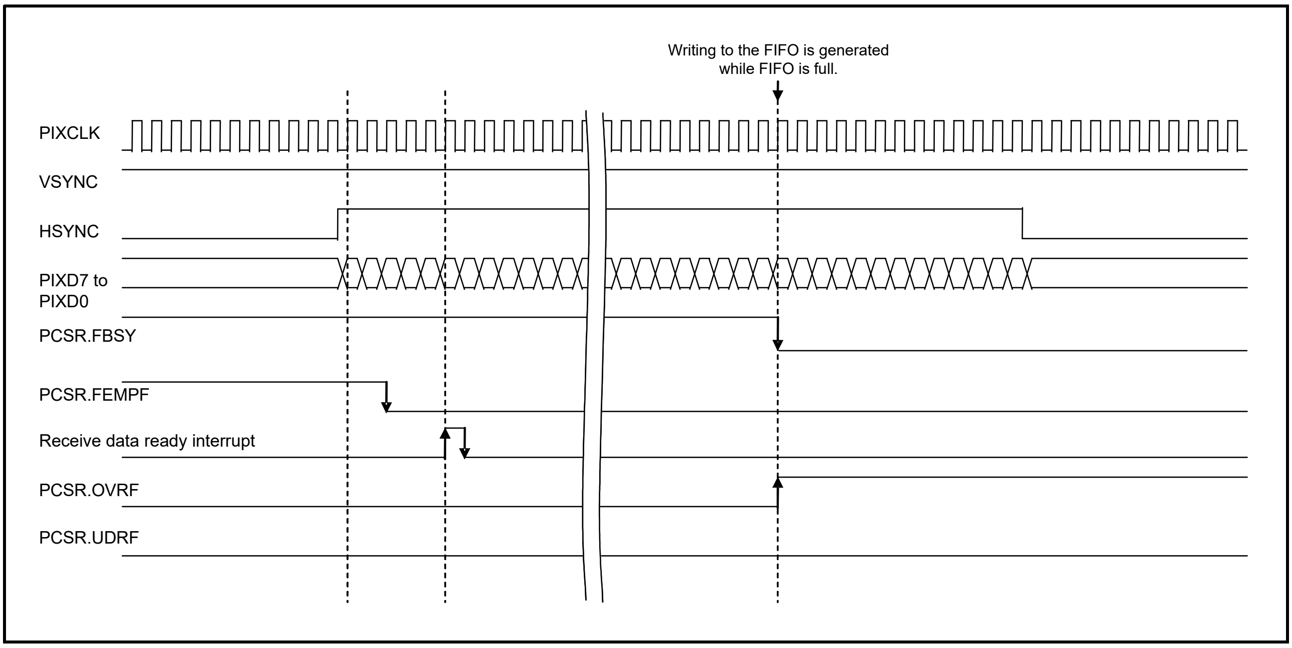


Figure 50.14 Overrun Detected

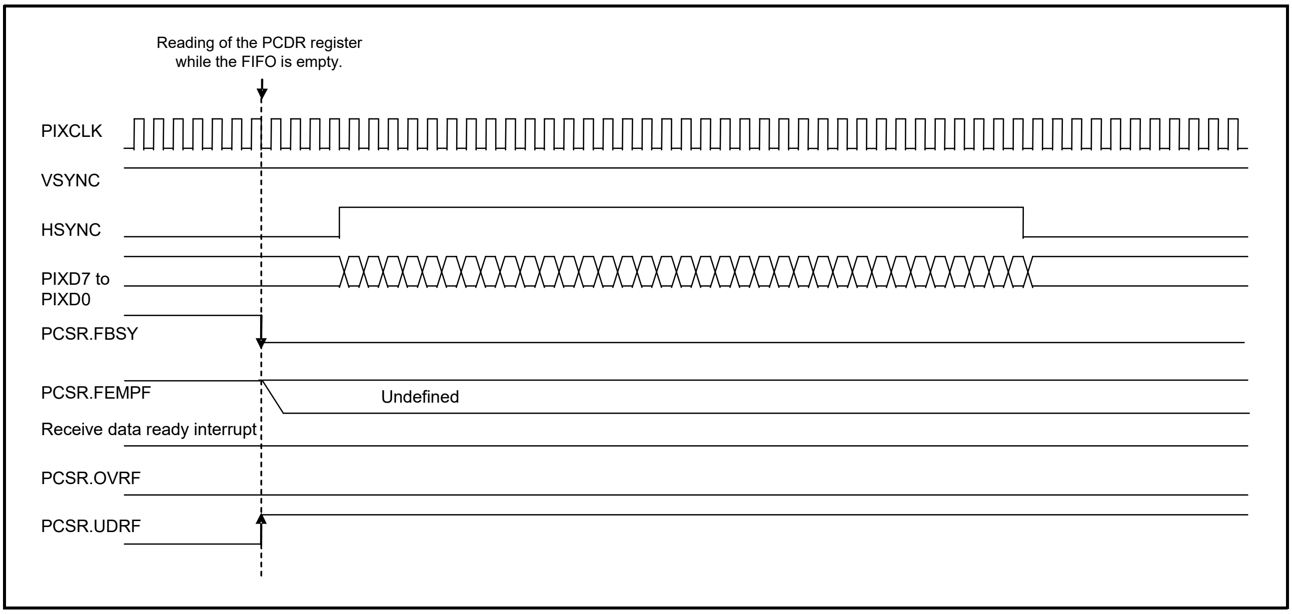


Figure 50.15 Underrun Detected

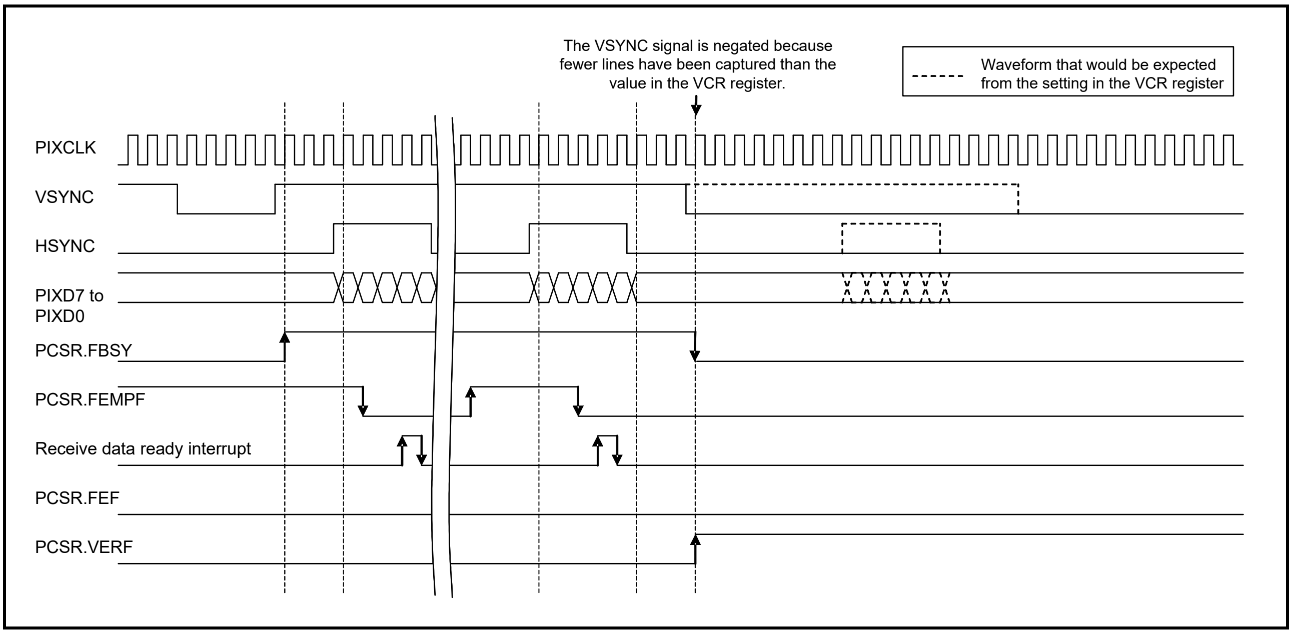


Figure 50.16 Vertical Line Number Setting Error Detected

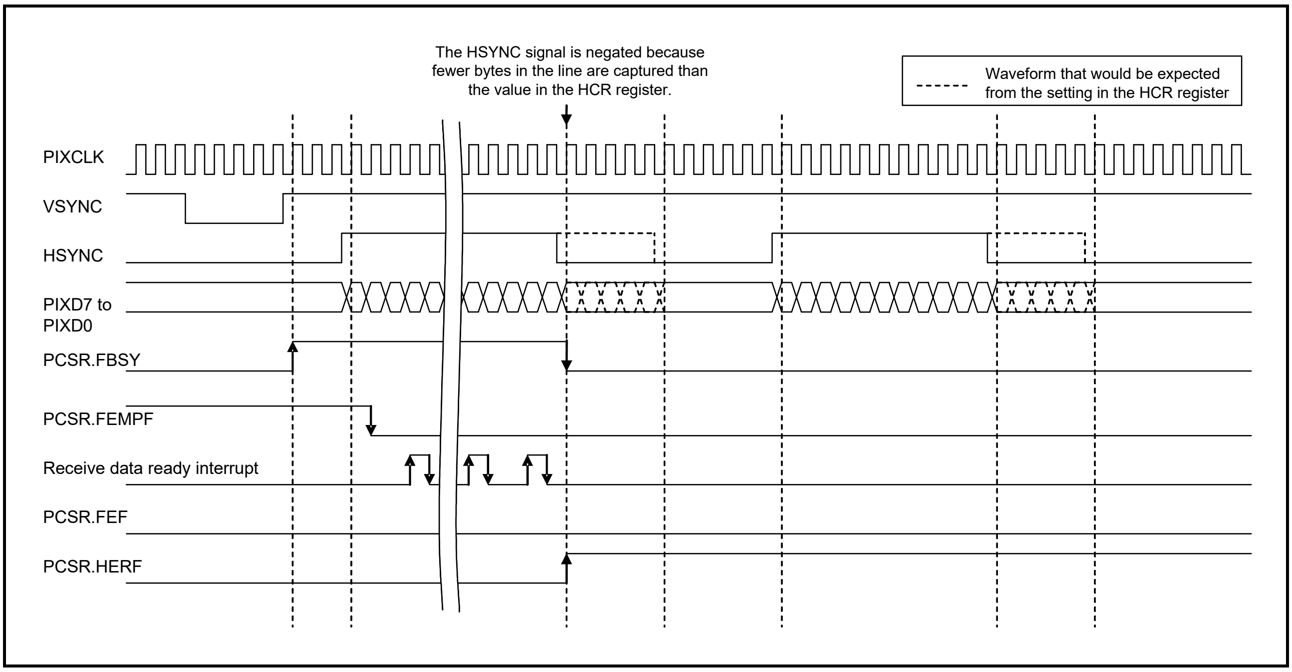


Figure 50.17 Horizontal Byte Number Setting Error Detected

50.3.8 Initial Settings

Figure 50.18 is a sample flowchart for initial settings. For a description of how to set up the input and output ports and the interrupt controller, refer to the descriptions given in the sections on the relevant blocks.

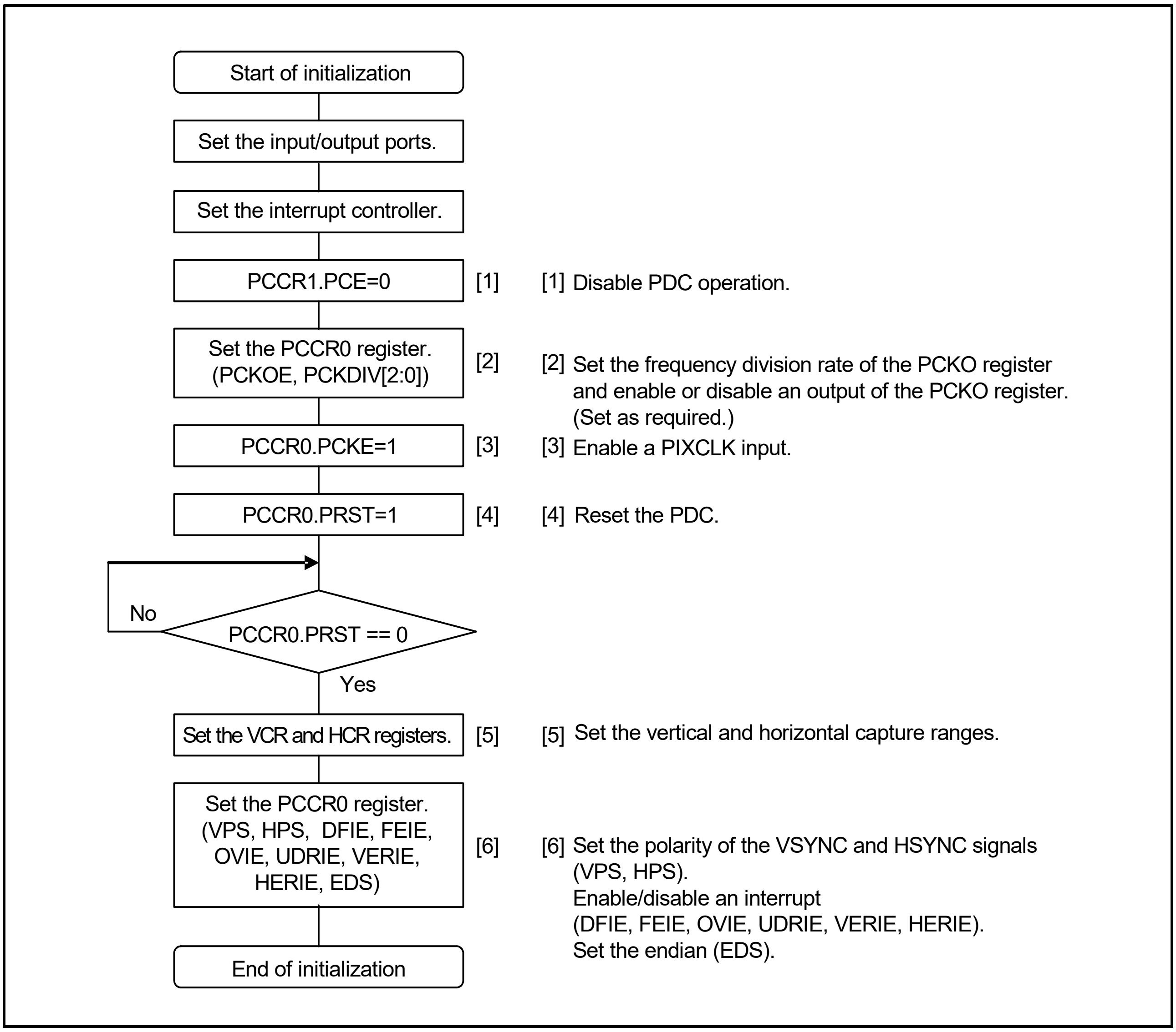
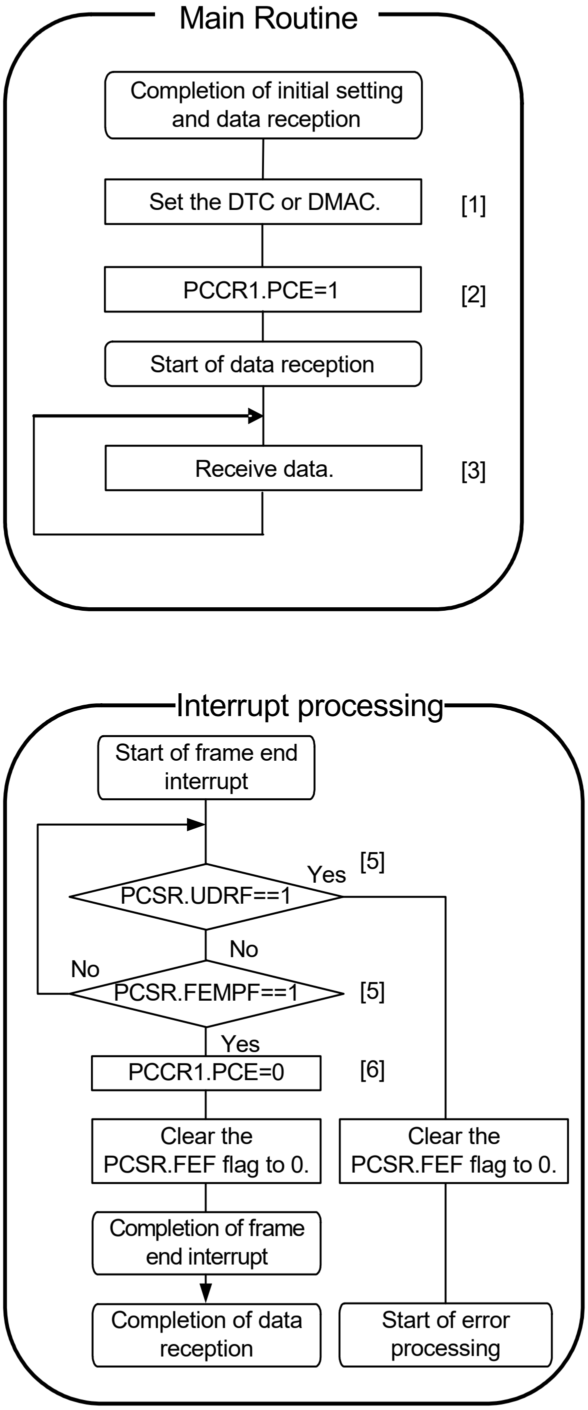
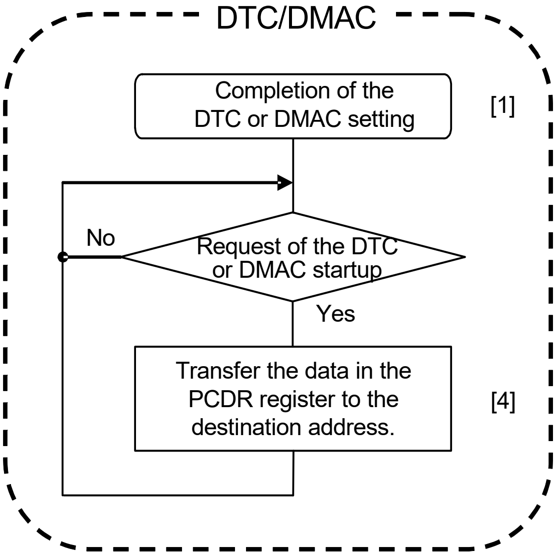


Figure 50.18 Sample Flowchart for Initial Setting of the PDC

50.3.9 Flows of Operations

Figure 50.19 shows an example of the flows of operations when the receive data ready interrupt (to start up the DTC or DMAC) and frame end interrupt are in use. For a description of how to set up the DTC or DMAC, refer to section 18, DMA Controller (DMACAa) and section 20, Data Transfer Controller (DTCb).





[1] Setting of the DTC or DMAC:
Set the DTC or DMAC according to the size of the captured image.

[2] Enabling of PDC operation:
Set the PCE bit in the PCCR1 register to 1 to enable PDC operation.

[3] Data reception:
Receive the data from the external I/O and wait for generation of a receive data ready interrupt (DTC/DMAC startup) or a frame end interrupt.

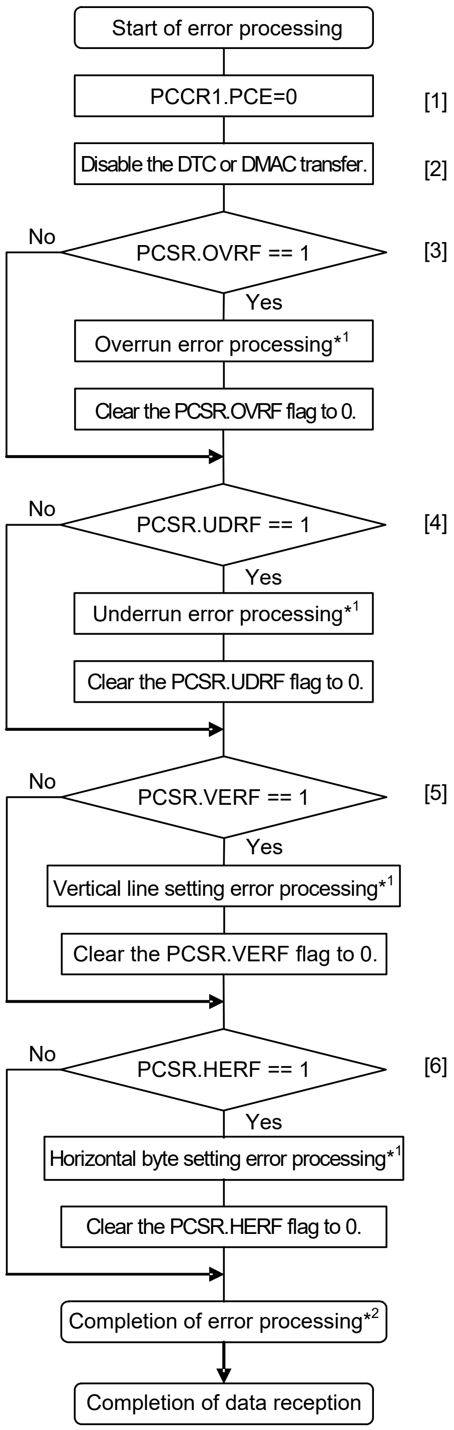
[4] Transfer by the DTC or DMAC:
Transfer the data in the PCDR register to the destination address in response to a receive data ready interrupt (DTC/DMAC startup).
Read the PCDR register in longword access and transfer 32 bytes of data per receive data ready interrupt.

[5] Frame end interrupt:
Read the FEMPF flag in the PCSR register and verify completion of the DTC or DMAC transfer.
If an underrun occurs during DTC or DMAC transfer, setting of the FEMPF flag may not proceed. When an underrun occurs, clear the FEF flag to 0 and branch to the appropriate error processing routine.

[6] Completion of data reception:
After completion of the transfer, clear the PCE bit in the PCCR1 register to 0, and then clear the FEF flag in the PCSR register to 0. This completes handling of the frame end interrupt and data reception.

Figure 50.19 Example of Flows of Operations

Figure 50.20 shows an example of the flow of operations in response to an error interrupt.



[1] Disabling of PDC operation:
Clear the PCE bit in the PCCR1 register to 0 to disable PDC operation.

[2] Disabling of the DTC or DMAC transfer:
Disable the transfer of the DTC or DMAC.

[3] Checking the overrun flag:
Read the PCSR register to check the state of the OVRF flag. If the flag is set, processing to handle the overrun error proceeds and the flag is then cleared.

[4] Checking the underrun flag:
Read the PCSR register to check the state of the UDRF flag. If the flag is set, processing to handle the underrun error proceeds and the flag is then cleared.

[5] Checking the vertical line setting error flag:
Read the PCSR register to check the state of the VERF flag. If the flag is set, processing to handle the vertical line setting error proceeds and the flag is then cleared.

[6] Checking the horizontal byte setting error flag:
Read the PCSR register to check the status of the HERF flag. If the flag is set, processing to handle the horizontal byte setting error proceeds and the flag is then cleared.

Note 1. Processing for each type of error is defined by the user, so the user needs to prepare error processing for each.

Note 2. If the CPU, DTC, or DMAC reads the PCDR register after an error interrupt has occurred in response to an interrupt source other than an underrun, an underrun may occur. If an underrun occurs on completion of error processing, run the error processing again and clear the UDRF flag. Since an underrun after error processing still requires detection, do not prohibit the underrun interrupt before the error processing.

Figure 50.20 Example of Error Processing Flow

50.3.10 Interrupt Source

The interrupt sources of the PDC are receive data ready, frame end, overrun, underrun, vertical line number setting error, and horizontal byte number setting error. The PDC can start up the DTC or DMAC in response to a receive data ready interrupt request for the transfer of data.

Table 50.5 summarizes the interrupt sources of the PDC. When an interrupt condition listed in Table 50.5 is satisfied, the corresponding interrupt is generated. In the case of receive data ready, the interrupt source flag is cleared by reading of the PCDR register. For the interrupt source flag for frame end, the FEF flag in the PCSR register is cleared. For an overrun, underrun, vertical line number setting error, or a horizontal byte number setting error, the flags must be checked to identify the error source flag because their interrupt vectors are allocated to the same address by the PCERI. Once identified, the corresponding error interrupt source flag (the OVRF, UDRF, VERF, or HERF flag) in the PCSR register should be cleared.

When the DTC or DMAC is to handle data transfer, whichever is selected should be set up first. After enabling the block to handle transfer, set up the PDC. For descriptions of how to set up the DTC and DMAC, refer to section 18, DMA Controller (DMACAa) and section 20, Data Transfer Controller (DTCb).

In addition, for receive data ready, if the ICU.IRn.IR flag is 1, the interrupt request signal is retained internally but an interrupt request is not output to the ICU, even if the interrupt condition is satisfied. In this case, however, only one interrupt request signal can be retained internally. If the ICU.IRn.IR flag is 0, the retained interrupt request signal is output to the ICU. After completion of output, the request flag is automatically cleared. An interrupt request signal retained internally can also be cleared by setting the corresponding interrupt enable bit (the DFIE bit in the PCCR0 register) to 0.

When the receive data ready interrupt is disabled from the enabled state, set the ICU.IRn.IR flag to 0 according to the following procedure.

1. Set the IERn.IENj bit to the value that disables interrupt requests.
2. Set the receive data ready interrupt enable bit (the DFIE bit) to "disabled" and read out the register after writing to verify that the bit has the new value.
3. Set the ICU.IRn.IR flag to 0.

Table 50.5 PDC Interrupt Source

Interrupt Source	Abbre-viation	Interrupt Conditions	DTC/DMAC Startup
Receive Data Ready	PCDFI	Receive data ready occurring while the DFIE bit in the PCCR0 register is 1.	Possible
Frame End	PCFEI	Frame end occurring while the FEIE bit in the PCCR0 register is 1.	Impossible
Errors	PCERI	An overrun occurring while the OVIE bit in the PCCR0 register is 1. An underrun occurring while the UDRIE bit in the PCCR0 register is 1. A vertical line number setting error occurring while the VERIE bit in the PCCR0 register is 1. A horizontal byte number setting error occurring while the HERIE bit in the PCCR0 register is 1.	Impossible

50.3.11 Reset State

Resetting of the PDC is divided into two types, PDC reset and other reset. Other resets include RES# pin reset, power-on reset, voltage monitoring reset 0, voltage monitoring reset 1, voltage monitoring reset 2, deep software standby reset, independent watchdog timer reset, watchdog timer reset, and software reset. Table 50.6 lists the range and states following the two types of reset.

Table 50.6 Reset State

	PDC Reset	Other reset
PCCR0	Retained	Reset
PCCR1	Retained	Reset
PCSR	Reset	Reset
PCMONR	Retained	Reset
PCDR	Retained	Reset
VCR	Retained	Reset
HCR	Retained	Reset

50.4 Usage Notes

50.4.1 Setting of the Module Stop Function

The module stop control register B (MSTPCRB) enables or disables operation of the PDC. Operation of the PDC is stopped when the value is that following a reset. The register becomes accessible following release from the module stop state. For details, refer to section 11, Low Power Consumption.

50.4.2 Notes on the Power Consumption Reduction Function

When power consumption by the PDC is to be reduced by using the power consumption reduction function, set the PCE bit in the PCCR1 register to 0 to disable operations for reception, and set the PCKE bit in the PCCR0 register to 0 to disable input through the PIXCLK pin. After that, use the power consumption reduction function.
If the setting of the PCKOE bit in the PCCR0 register is 1, it should be set to 0 to stop output of the PCKO signal, and input through the PIXCLK pin should be disabled in the way described above. After that, use the power consumption reduction function.
On return from the low power consumption state, set the PCKE bit to 1, and after enabling input through the PIXCLK pin, use the PRST bit to initialize the PDC.

50.4.3 Notes on Error Interrupts

When an error interrupt occurs, there is a possibility that the DTC or DMAC might be transmitting parallel data depending on their operation state. Therefore, transmitting executed by the DTC or DMAC should be prohibited after the PDC operation is prohibited at the top of the error interrupt processing routine (PCCR1.PCE = 0).

50.4.4 Notes on Use of the DTC

When the DTC is used with the receive data ready interrupt, set the DISEL bit in the MRB register to 0 and the SZ bit in the MRA register to 10b.
The maximum number of blocks the DTC can transfer in block transfer mode is 65536. If 32 bytes are transferred per block transfer, this represents a total of up to 2097152 bytes. If more data are to be transferred, set up the DTC again during the horizontal blanking period.

50.4.5 Notes on Use of the DMAC

When the DMAC is used with the receive data ready interrupt, set the DISEL bit in the DMCSL register to 0, set the SZ bit in the DMTMD register to 10b, and set the IPRn.IPR[3:0] bit responding to an ICU receive data ready interrupt to level 0 (the interrupt is prohibited).
The maximum number of blocks the DMAC can transfer in block transfer mode is 65536. If 32 bytes are transferred per block transfer, this represents a total of up to 2097152 bytes. If more data are to be transferred, set up the DMAC again during the horizontal blanking period.

50.4.6 Notes on Start of Transfer

When transfer starts while the ICU.IRn.IR flag is 1, an interrupt request signal is internally retained after the start of transfer. This may cause the ICU.IRn.IR flag to behave in an unexpected way.
In such cases, therefore, clear the interrupt request flag by following the procedures below before enabling operations (i.e. before setting the PCE bit in the PCCR1 to 1).

(1) Ensure that transfer is stopped (the PCE bit in the PCCR1 is 0).
(2) Set the corresponding interrupt enable bit (the DFIE bit in the PCCR0 register) to 0.
(3) Read out the corresponding interrupt enable bit (the DFIE bit in the PCCR0 register) to ensure that it is 0.
(4) Set the ICU.IRn.IR flag to 0.